

CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE · INDIA

AUTOMOBILE INDUSTRY PROJECT

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Under the guidance of
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A Project report submitted in partial fulfillment of the requirements for the award of degree of Master of Computer Applications of

CHRIST(Deemed to be University)

May - 2025

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CERTIFICATE

This is to certify that the report titled ***"Automobile Industry Project"*** *is a bonafide record of work done by* ***Shrey Gupta(2347154)*** *of CHRIST(Deemed to be University), Bangalore, in partial fulfillment of the requirements of VI Trimester MCA during the year 2024-25.*

Head of the Department

Project Guide

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Shrey Gupta

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Private & Confidential

April 04, 2025

To
The Head of the Department
Department of Computer Science
Christ University
Bangalore, India

Dear HOD,

Subject: Confirmation of Full-time Employment and Industry Project Participation

This is to formally confirm that **Shrey Gupta**, a graduate of Christ University, has joined Ipsos as a full-time 'Analyst' in our Modelling team, effective **December 23, 2024**. Since joining, He has successfully completed all required training.

Furthermore, He is actively engaged in a live industry project at Ipsos, contributing to real world business challenges through data-driven insights and strategic analysis.

We appreciate the academic foundation provided by Christ University, which has equipped **Shrey Gupta** with the necessary skills to excel in their professional journey. Please feel free to reach out should you require any further details.

Sincerely,



Gaurav Ajmera
Associate Director - HR
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ABSTRACT

The **Marketing Optimization System for Automobile Client** is an advanced solution designed to enhance marketing budget allocation using artificial intelligence, predictive modeling, and data-driven insights. Traditional budget allocation methods often rely on manual decision-making, leading to inefficiencies, suboptimal spending, and missed opportunities. The **Marketing Mix Modeling (MMM) system**, a core component of this platform, addresses these challenges by leveraging AI-driven analytics to quantify the impact of various marketing channels, optimize budget distribution, and maximize return on investment (ROI). By seamlessly integrating data processing, statistical modeling, and scenario-based simulations, this project ensures an optimized and strategic approach to marketing expenditure.

The system is designed for marketing analysts, business strategists, and financial planners aiming to improve budget efficiency and enhance sales performance. It benefits stakeholders by identifying high-performing marketing channels, balancing base and incremental contributions, and adapting to dynamic market trends. Key features such as **predictive budget allocation, performance attribution modeling, and competitor benchmarking** make it particularly useful for brands seeking data-backed decision-making. Additionally, it enables scenario analysis, allowing teams to explore various budget distribution strategies and assess their potential outcomes in real-time.

The development of the **AI-Driven Marketing Optimization System** is underpinned by a robust AI-powered methodology, integrating multiple technologies. SQL and data extraction tools facilitate historical data collection, while advanced econometric models such as **Multi-Touch Attribution (MTA) and Marketing Mix Modeling (MMM)** analyze the impact of marketing investments. The platform is built using **Python, Excel, and SQL-based reporting tools** for data processing and insights generation. AI frameworks such as **scikit-learn and TensorFlow** enable predictive analytics, while visualization tools like **Tableau/Power BI** provide actionable insights through interactive dashboards.

The AI-Driven Marketing Optimization System introduces an **intelligent and data-centric approach** to marketing strategy, offering **enhanced budget efficiency, real-time insights, and predictive scenario planning**. With its advanced analytical capabilities, the system empowers Automobile Client's marketing team to optimize spending, improve sales performance, and drive sustainable business growth.

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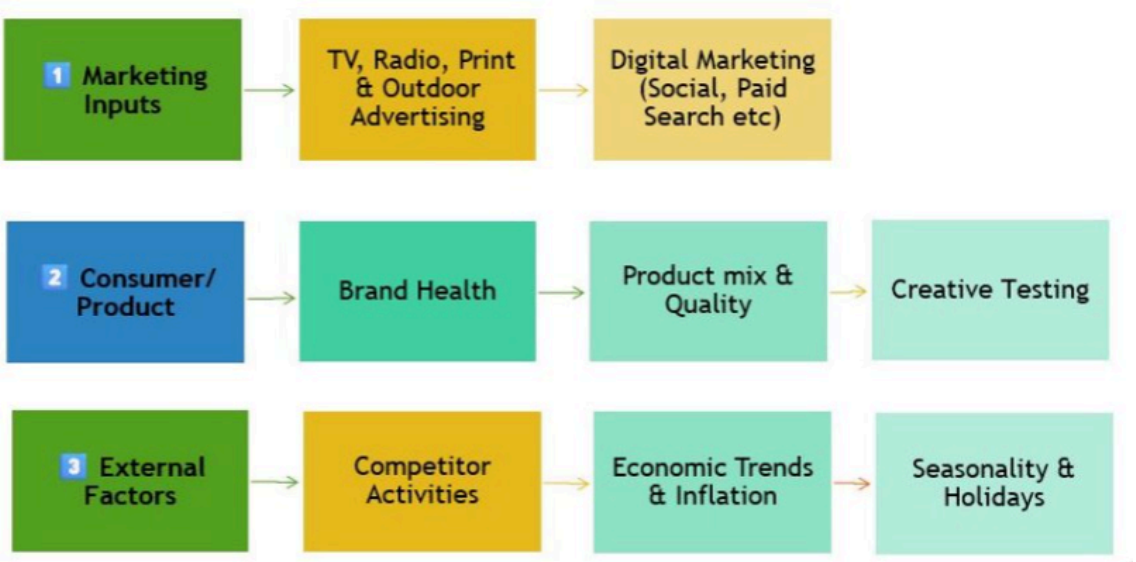
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INTRODUCTION

In today's highly competitive automotive market, effective marketing plays a crucial role in influencing consumer decisions and driving vehicle sales. With increasing complexities in customer behavior, digital transformation, and ever-changing market dynamics, companies

like Automobile must constantly innovate and refine their marketing strategies to remain ahead of the competition. One of the most critical aContribution Filets of this strategy is the **allocation of the marketing budget** across various channels to maximize visibility, engagement, and ultimately, vehicle sales.

This chapter introduces the project undertaken to assist **Automobile Car Corporation** in enhancing its marketing strategy by optimizing the allocation of its marketing budget. The chapter lays the groundwork by explaining the current scenario, the challenges faced with traditional marketing practices, and the necessity to move towards a **data-driven and predictive approach** for budget distribution.



Background

Automobile, as a renowned automobile manufacturer, invests a significant portion of its operational resources in marketing activities, including **organic search (SEO)**, **paid social media campaigns**, and **competitive benchmarking** to maintain its brand presence and customer engagement. However, the company's existing approach relies heavily on conventional budget allocation methods, which are largely based on **historical performance, intuition, or fragmented insights**. While this approach may have worked reasonably well in the past, it lacks the precision required to adapt to today's fast-paced and digitally influenced consumer landscape.

Global trends indicate that modern businesses are increasingly adopting **data-driven strategies** to fine-tune their marketing operations. Leveraging advanced analytics and predictive models allows businesses to gain actionable insights and make **informed decisions** that result in **higher returns on investment (ROI)**. This project is aligned with such a forward-thinking strategy, aiming to enable Automobile to allocate its marketing budget more scientifically and effectively.

1.1 PROJECT DESCRIPTION

Automobile is globally recognized as a leading premium automobile manufacturer, well-known for its focus on safety, innovation, and sustainability. As a major player in the automotive sector, Automobile has consistently invested substantial resources into marketing activities to maintain and strengthen its brand presence across key markets worldwide.

To support its **vehicle sales objectives**, Automobile engages in a variety of marketing initiatives, including:

- **Organic Search (SEO):** Automobile invests in search engine optimization to enhance the visibility of its website and product pages on search engines like Google and Bing. The goal is to attract potential customers organically when they search for terms related to vehicles, safety features, or automotive technology.

- **Paid Social Media Campaigns:** Automobile actively utilizes paid advertisements on social media platforms such as Facebook, Instagram, LinkedIn, and Twitter to promote its latest vehicle models, highlight brand values, and target Contribution Fileific customer segments with tailored content.

- **Competitive Benchmarking:** Automobile continually monitors its competitors' marketing strategies, pricing tactics, and promotional campaigns. Insights from these activities are crucial in positioning Automobile's offerings relative to competitors like BMW, Audi, and Mercedes-Benz.

Despite these efforts, **Automobile's traditional marketing budget allocation process** still has significant limitations.

The Conventional Approach

Historically, Automobile's marketing budget allocation decisions have been based on a combination of:

- **Past performance metrics** (e.g., previous campaign results).

- **Managerial intuition and experience** within regional and global marketing teams.

- **Fragmented data** coming from various sources and marketing channels that are not always integrated or centrally analyzed.

While these traditional methods have helped the company maintain a competitive position over the years, they tend to lack the sophistication necessary to drive **continuous improvement and optimization**. The complexity of the modern automotive market, coupled

with shifts in customer behavior and technological disruption, means that **intuition-driven decisions alone are no longer sufficient**.

1.2 EXISTING SYSTEMS

The challenges faced by Automobile in its current system include:

1. **Siloed Marketing Channels:** Each channel (e.g., SEO, social media, display DATASET) may operate independently, leading to inefficiencies due to poor cross-channel coordination.

2. **Reactive Planning:** The absence of predictive tools forces marketing teams to adopt a reactive stance, only adjusting budget allocations after campaigns have concluded and results are analyzed.

3. **Subjective Decision-Making:** Without a unified analytical framework, marketing executives often rely on personal judgment or static allocation rules, which may not reflect the true market dynamics.

4. **Limited Agility:** In fast-changing markets, where customer preferences, competitor tactics, and external economic factors can shift rapidly, Automobile's current system does not offer the agility needed to reallocate budgets in real-time.

The Need for Change

The rapid **digitalization of marketing ecosystems**, fueled by big data, machine learning, and advanced analytics, has fundamentally changed how organizations optimize their marketing investments. Global leaders in multiple industries are moving towards **data-driven frameworks** that allow them to:

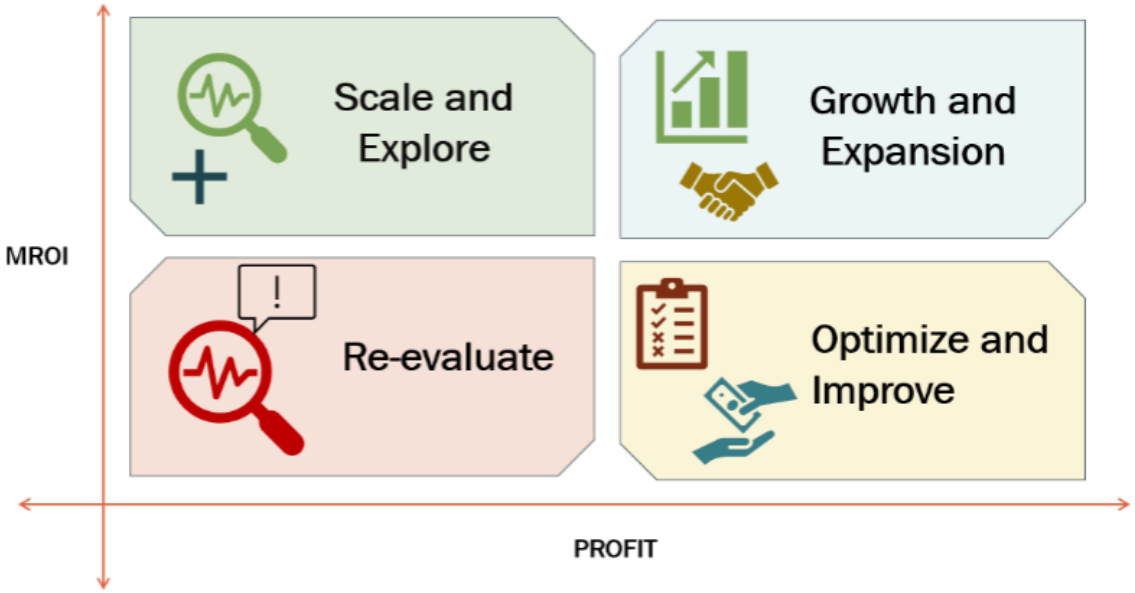
- Forecast campaign performance.

- Predict customer responses to marketing stimuli.

- Allocate budgets to channels with the **highest marginal returns**.

- Simulate "what-if" scenarios to understand the potential impact of changes in spending.

In this competitive environment, companies that embrace **data-driven marketing strategies** stand to gain a considerable edge in terms of cost-efficiency, brand loyalty, and sales uplift. Automobile recognizes this shift and aims to be at the forefront of this transformation.



Alignment with Modern Business Trends

Globally, leading firms in the automotive sector and beyond are increasingly embedding **data science and AI capabilities** into their marketing operations. By leveraging predictive analytics, businesses can:

- Uncover patterns in customer behavior across channels and touchpoints.
- Quantify the impact of seasonal trends, macroeconomic indicators, and competitive pressures on campaign performance.
- Optimize marketing spend to achieve maximum **ROI** while staying within budget constraints.

This project aims to place Automobile firmly within this paradigm by creating a solution that **systematically analyzes historical marketing data**, **predicts vehicle sales outcomes**, and **optimizes marketing budgets** based on real-world variables.

1.3 OBJECTIVES

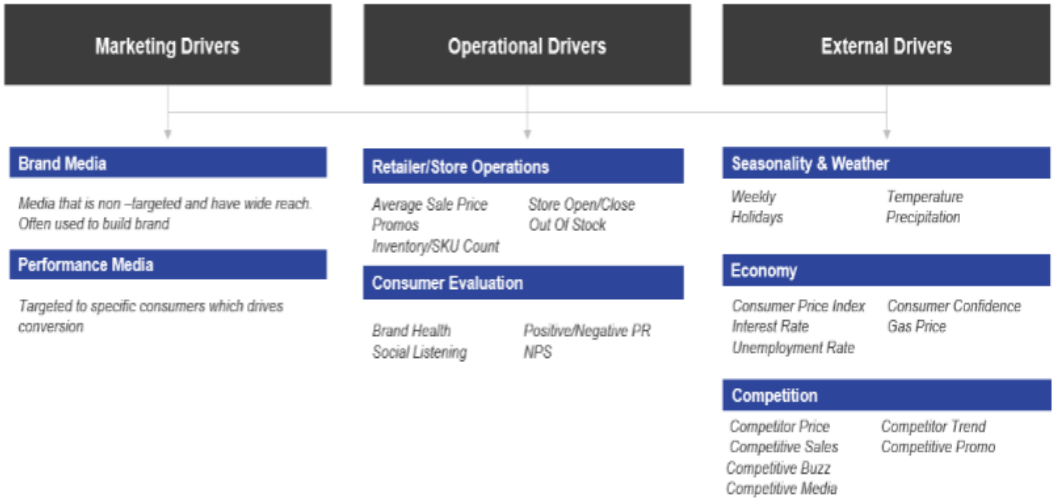
For Automobile, this shift toward a **predictive and data-driven marketing strategy** is not just a technical upgrade but a **strategic necessity**. Key reasons include:

- **Sustaining Brand Leadership:** In an industry where safety, sustainability, and innovation are highly valued, Automobile's marketing needs to be as precise and

impactful as its product offerings.

- **Adapting to Changing Customer Journeys:** Modern consumers interact with brands across multiple digital and offline touchpoints before making a purchase decision. Understanding these journeys requires deeper data insights.

- **Maximizing Marketing ROI:** As Automobile continues to expand into new markets and launch new models (such as its electric vehicle range), efficiently allocating marketing dollars will be essential to achieving sustainable growth and profitability.



Need for Marketing Budget Optimization

The traditional process of allocating marketing budgets often suffers from several drawbacks:

- **Subjectivity and Bias:** Decision-makers often rely on personal experience or gut feeling, which can lead to inconsistent and sub-optimal outcomes.

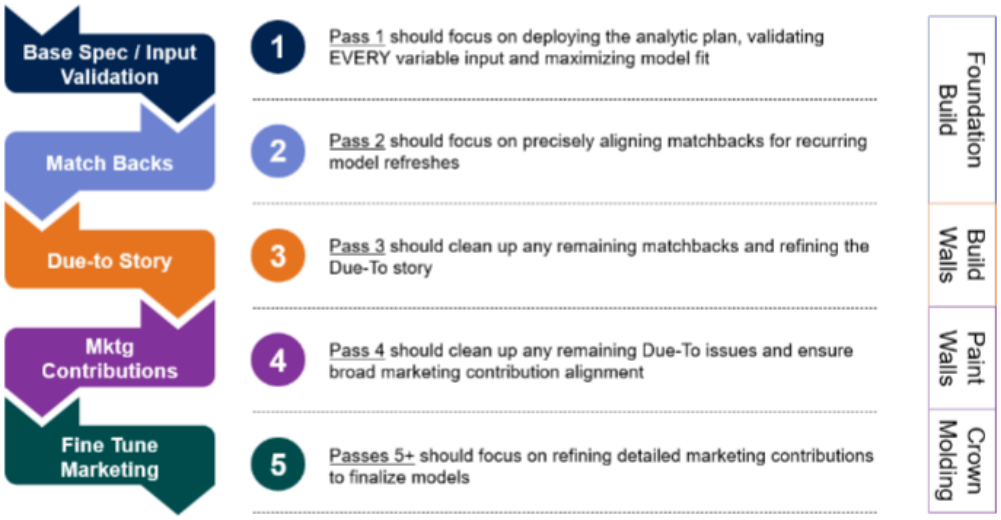
- **Fragmented Data Sources:** Different marketing channels operate in silos, making it difficult to get a holistic view of marketing effectiveness across the entire ecosystem.

- **Lack of Predictive Insights:** Without a predictive framework, it becomes challenging to forecast the impact of marketing spend on vehicle sales.

- **Dynamic Market Conditions:** Consumer behavior, competitor strategies, and macroeconomic trends are constantly changing, necessitating a flexible and responsive budget allocation strategy.

1.4 PURPOSE SCOPE AND APPLICABLITY

Given these challenges, there is an urgent need to move towards a **systematic, data-driven approach** that not only considers historical performance but also leverages predictive analytics to suggest **optimal spending decisions**.



Target Users

The system being developed for Automobile's marketing budget optimization is designed to serve a diverse range of internal stakeholders, each with Contribution Fileific needs and responsibilities. The system's insights will be crucial in supporting data-driven decision-making across multiple functional units. Below is an in-depth breakdown of the target users and how each will leverage the system:

1. Marketing Teams

Automobile's marketing teams, both at the regional and global levels, are at the forefront of executing campaigns that drive customer engagement and brand awareness. These teams manage various channels, such as SEO, paid social, digital DATASET, and offline initiatives.

How the system will help:

- **Optimized Campaign Planning:** The system will provide actionable insights into which marketing channels, time periods, and customer segments yield the best return on investment (ROI). This will enable marketers to allocate resources to high-performing initiatives and reduce waste on underperforming campaigns.

- **Channel Effectiveness Insights:** The system will highlight how individual marketing channels (e.g., organic search, paid social, or display DATASET) contribute to Automobile's vehicle sales and brand metrics.

- **Real-time Recommendations:** The predictive model will allow marketing teams to make agile, data-backed adjustments to ongoing campaigns, improving responsiveness to market dynamics.

Benefits:

- Enhanced campaign efficiency and effectiveness.

- Reduced reliance on guesswork or anecdotal evidence.

- Increased ability to meet key performance indicators (KPIs) such as lead generation, conversion rates, and customer engagement.

2. Business Analysts

Business analysts within Automobile are responsible for interpreting complex data sets, identifying trends, and providing management with reports that inform tactical and strategic decisions. Analysts are often the bridge between raw data and business decision-makers.

How the system will help:

- **Deeper Analytical Insights:** The system will provide a comprehensive view of historical and real-time marketing data, including channel performance, customer behavior, and external influencing factors (e.g., competitor activity, seasonality).

- **Data Visualization:** The platform will include dashboards, charts, and visual analytics tools to help analysts easily interpret data and trends.

- **Scenario Analysis:** Business analysts will be able to use the system to run simulations and "what-if" analyses to determine how budget reallocations could impact vehicle sales outcomes.

Benefits:

- Improved ability to create insightful, data-driven reports.
- Stronger narratives and recommendations for stakeholders based on solid analytics.
- More accurate forecasting of marketing-driven sales impacts.

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3. Financial Planners

Financial planners are tasked with managing Automobile's broader financial resources, ensuring that marketing budgets are aligned with corporate fiscal policies, sales targets, and long-term financial goals.

How the system will help:

- **Budget Optimization Recommendations:** The system will generate optimized budget allocation strategies across multiple channels and campaigns to ensure that marketing spend aligns with financial objectives such as ROI targets and overall profitability.
- **Cost-Benefit Analysis:** Financial planners can leverage insights to understand how every marketing dollar contributes to sales and brand equity.
- **Resource Allocation Alignment:** The system will support integrated planning by aligning marketing budgets with corporate financial frameworks and forecasts.

Benefits:

- Improved budget planning and allocation processes.
- Enhanced financial discipline in marketing expenditure.
- Greater accountability through data-supported budget justifications.

4. Senior Management

Senior management, including executives such as the Chief Marketing Officer (CMO), Chief Financial Officer (CFO), and regional business heDATASET, are responsible for shaping Automobile's strategic direction and ensuring the company meets its long-term objectives.

How the system will help:

- **High-level Decision Support:** The system will distill complex data into clear and concise insights, helping senior leaders make informed decisions regarding resource allocation, market expansion, and competitive positioning.

- **Strategic ROI Forecasting:** Executives will be able to see projections on how different budget allocation strategies influence vehicle sales, brand reach, and profitability.

- **Macro and Micro Analysis:** Senior leaders will benefit from both the big picture and granular insights, providing the flexibility to make decisions at both the regional and global levels.

Benefits:

- Enhanced ability to steer marketing strategy in line with corporate goals.

- Increased confidence in data-driven decisions, reducing dependency on gut-feeling or traditional practices.

- Improved communication between departments through transparent, quantifiable metrics.

1.5 OVERVIEW OF THE PROJECT

Automobile, like many legacy automotive companies, currently operates under a **traditional marketing budget allocation system** that is primarily driven by historical data, business intuition, and siloed insights. While this approach has delivered reasonable results over the years, it is increasingly becoming less effective in the fast-evolving digital landscape where customer behaviors and market dynamics shift rapidly.

1. Dependence on Historical Trends

The existing system primarily uses historical spending patterns and prior campaign performance as the basis for determining future budget allocations. For example:

- If a Contribution Fileific channel, such as paid social media, performed moderately well in the previous quarter or year, a similar or slightly adjusted budget is allocated to it for the upcoming period.

- This “trend-following” approach does not account for recent changes in consumer behavior, competitor activity, or macroeconomic variables (e.g., inflation, supply

chain disruptions) that may have altered the effectiveness of marketing channels.

2. Intuition and Heuristic-Based Decision Making

Much of the marketing budget planning is also influenced by the experience and intuition of marketing managers or teams. While experienced professionals bring valuable insights, this heuristic-based approach introduces subjectivity and lacks the consistency that data-driven methodologies provide.

Common issues include:

- **Biases:** Personal preferences or past experiences of decision-makers may favor certain channels over others without a solid data-backed rationale.

- **Inflexibility:** Decisions made early in the fiscal year often remain unchanged, even if mid-year performance metrics indicate that some channels are underperforming.

3. Fragmented Insights Across Channels

The current system operates in silos, with each marketing channel (e.g., organic search, paid social, offline campaigns) often being evaluated independently rather than as part of an integrated ecosystem. Some of the challenges include:

- **Lack of cross-channel attribution modeling:** There is limited understanding of how multiple touchpoints work together to influence a customer's purchase decision. For instance, a customer may first encounter a Automobile ad on social media but later convert after seeing an organic search result, yet these touchpoints are not effectively linked.

- **Isolated data sets:** Each team or department may use different tools and metrics, leading to inconsistencies in how performance is measured and reported.

4. Absence of Predictive or Optimization Tools

The current system lacks advanced analytical or predictive capabilities, such as machine learning models or optimization algorithms. As a result:

- There is **no scientific method** to predict the ROI of future marketing investments based on past trends, competitive landscape shifts, or seasonal variations.

- Budget recommendations are **static**, often based on "fixed percentage" rules (e.g., allocate 30% to digital, 40% to TV, etc.) without assessing the real-time effectiveness

of each channel.

5. Limited Responsiveness to Market Dynamics

Due to its rigid and manual nature, the system struggles to react promptly to sudden changes such as:

- **Competitor actions:** If a competitor ramps up spending in paid search or launches aggressive promotions, the existing system does not provide timely recommendations to counteract these moves.

- **Seasonal or regional shifts:** Some channels may perform better in certain regions or seasons (e.g., winter promotions for SUVs), but the current system does not dynamically reallocate budgets based on these patterns.

6. Impact on Business Outcomes

Because of the above limitations, Automobile's current approach to marketing budget allocation may result in:

- **Suboptimal ROI:** Funds may be allocated to channels that yield lower sales impact, while high-performing channels may remain underfunded.

- **Missed opportunities:** Without predictive insights, Automobile may miss trends or emerging channels that could outperform traditional ones.

- **Inefficient resource utilization:** The marketing spend may not fully align with broader business and financial goals, such as boosting sales for Contribution Fileific models (e.g., XC40, XC60) or capturing market share in key regions.

SYSTEM ANALYSIS AND REQUIREMENTS

2.1 PROBLEM DEFINITION

Automobile's marketing department currently relies heavily on **traditional and intuition-based methods** to distribute its marketing budget across various channels (e.g., digital, print, television, and outdoor advertising). While this approach has served the company in the past, the modern marketing landscape has evolved significantly due to rapid digitalization, increasing customer expectations, and the availability of sophisticated data analytics tools.

Without leveraging a **data-driven strategy**, Automobile's marketing decisions are based on general assumptions and past experiences rather than empirical evidence. As a result, several critical issues have emerged that are negatively impacting both operational effectiveness and the company's bottom line:

1. Overspending on Underperforming Channels

Automobile continues to allocate significant portions of its marketing budget to certain channels based on historical precedence or management preferences. However, without data-backed insights, there is little evidence to confirm that these channels are currently yielding strong returns. This results in:

- **Wasted expenditure** on channels that no longer effectively reach or convert the target audience.
- **Opportunity cost**, where funds that could have been used in more productive channels are locked into low-performing ones.
- **Inertia bias**, where decisions are based on "what has always been done," rather than adapting to changing market dynamics.

2. Insufficient Investment in High-Potential Channels

Conversely, emerging channels or segments that could be delivering higher ROI are often underfunded due to the lack of data supporting reallocation. This leDATASET to:

- **Missed growth opportunities** where promising platforms like Contribution Fileific social media networks, influencer marketing, or programmatic advertising remain underutilized.

- **Inability to capitalize on customer trends**, as new customer touchpoints are ignored in favor of conventional media.

- **Reduced agility** in adapting to dynamic customer behaviors and technological advancements in the digital marketing ecosystem.

3. Limited Visibility into Campaign Performance

Currently, Automobile's marketing team lacks a unified platform that consolidates insights from diverse campaigns across different segments and regions. This fragmented view results in:

- **Siloed data** that resides in separate tools and systems, making it hard to get a holistic picture of campaign effectiveness.

- **Inconsistent KPIs and metrics**, where different teams may be evaluating success differently across campaigns.

- **Challenges in customer segmentation**, as insights on how different customer profiles respond to various campaigns are not fully utilized.

- **Difficulty identifying synergies** between different marketing channels (e.g., how TV DATASET impact online search or social media engagement).

4. Poor Forecasting of Campaign Outcomes

Automobile's reliance on manual budget planning and subjective decision-making hinders its ability to accurately predict the outcomes of marketing campaigns. This problem is further exacerbated by:

- **Limited historical data utilization**, as past campaigns are not systematically analyzed to inform future ones.

- **Inability to simulate various budget allocation scenarios** to understand potential trade-offs between channels or customer segments.

- **Reactive planning**, where marketing decisions are made in response to past performance rather than proactively planning based on predictive models.

- **Sub-optimal marketing-mix modeling (MMM)** capabilities, where the cause-and-effect relationship between spend and sales is not quantified in a structured way.

Strategic Implications

Collectively, these challenges result in **reduced marketing efficiency** and **diminished overall profitability**. The lack of robust analytics makes it harder for Automobile's marketing team to justify investments, optimize their media mix, or align marketing strategies with overarching business objectives.

In today's data-centric world, where **consumer behaviors shift rapidly** and **competition is fierce**, Automobile is at risk of losing its competitive edge if it does not modernize its approach.

The Need for Change

To address these pressing concerns, Automobile requires a **systematic and analytical solution** that can:

- **Leverage data from multiple platforms** (e.g., CRM, media buying platforms, campaign reports).

- **Model and quantify the effectiveness of each marketing channel**.

- **Forecast campaign outcomes more accurately**.

- **Deliver actionable recommendations** for rebalancing and optimizing the marketing budget across regions, customer segments, and platforms.

The goal is to empower Automobile's marketing team to make **informed, evidence-based decisions** that will enhance customer targeting, improve conversion rates, and significantly increase marketing ROI.

Purpose of System Analysis

The purpose of this system analysis is to:

1. **Understand and document the existing marketing processes**, identifying inefficiencies and pain points.

2. **Determine business and technical requirements** to improve decision-making regarding budget allocation.

3. **Align the system's capabilities with Automobile's business goals**, such as improving customer engagement, reducing unnecessary spending, and driving higher marketing ROI.

4. Lay the groundwork for system design by clearly defining what the system will do, who will use it, and under what conditions.

2.1 REQUIREMENTS SPECIFICATIONS

System requirements are divided into two main categories: **Functional Requirements** and **Non-Functional Requirements**.

1. Functional Requirements

Functional requirements define what the system is expected to do, focusing on Contribution Fileific behaviors and functions. For this project, the system should:

- **Data Integration**: Collect and integrate data from various sources such as CRM systems, social media platforms, email marketing tools, and website analytics.

- **Customer Segmentation**: Analyze customer data and segment customers based on demographic, geographic, behavioral, and psychographic criteria.

- **Budget Optimization Algorithm**: Implement a machine learning or statistical model that will recommend optimal budget distribution across channels based on historical data, customer segments, and campaign performance.

- **Channel Performance Analysis**: Provide reports on the performance of each marketing channel (e.g., CPC, CTR, ROI per channel).

- **Forecasting Module**: Predict future campaign performance and expected returns based on different budget allocation scenarios.

- **User Dashboard**: Develop an intuitive dashboard that allows marketing managers to visualize insights, run simulations, and generate actionable reports.

- **Decision Support System (DSS)**: Assist users in making informed decisions by providing recommendations, alerts, and what-if analysis tools.

Non-Functional Requirements

While functional requirements describe the Contribution Fileific capabilities and features the system must deliver, **non-functional requirements (NFRs)** focus on the quality attributes of the system — how it should perform and behave under certain conditions. These requirements ensure that the system is not only functional but also efficient, secure, user-friendly, and capable of supporting the business both now and in the future.

For Automobile's marketing budget optimization system, the following non-functional requirements are critical:

1. Performance

The system must be capable of handling **large and complex datasets** originating from multiple marketing platforms (e.g., Data Repository, Marketview, SQL databases) with speed and efficiency.

- **Data Processing Efficiency:** The system should ingest, clean, process, and analyze datasets with minimal latency to enable swift decision-making.

- **Near Real-Time Insights:** Marketing teams often need to make rapid adjustments to campaigns based on market conditions. Therefore, the system should be able to generate budget recommendations and optimization strategies in **near real-time**.

- **Optimized Query Handling:** Given that marketing data can be highly granular (e.g., broken down by channel, region, and customer segment), the system must support fast execution of complex queries and modeling tasks without excessive resource consumption.

2. Scalability

As Automobile continues to grow and expand its marketing operations across regions and customer segments, the system must be able to **scale seamlessly** to handle:

- **Increased Data Volumes:** More customer data, campaign records, and channel metrics as the business and marketing footprint expand.

- **Higher User LoDATASET:** The possibility of more users (e.g., marketing analysts, managers) accessing and utilizing the system simultaneously.

- **Modular Expansion:** The system should support the addition of new features, channels, or integrations (e.g., emerging marketing platforms or external data sources) without requiring major architectural overhauls.

3. Security

Given the sensitivity of the data involved — including customer behavior, campaign performance, and financial records — **security is paramount**. The system must ensure:

- **Data Privacy:** All data handling and storage should comply with international data protection regulations, such as the **General Data Protection Regulation (GDPR)** and other relevant local regulations.

- **Access Controls:** Implement strict **role-based access control (RBAC)** to ensure that only authorized personnel can access sensitive information.

- **Encryption:** Employ end-to-end encryption, both at rest (when stored in databases) and in transit (when data is being transferred), to protect against data breaches.

- **Audit Trails:** Maintain detailed logs of all system interactions to detect and prevent unauthorized activities.

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4. Usability

The system should be **intuitive and accessible** to marketing professionals who may have varying levels of technical expertise:

- **User Interface (UI):** Develop a **clean and easy-to-navigate dashboard** where users can view key insights, recommendations, and reports without requiring deep technical knowledge.

- **User Experience (UX):** Optimize workflows so that marketing managers can easily run scenario analyses, generate reports, and apply recommendations with minimal training.

- **Guided Features:** Include in-system tooltips, documentation, and tutorials to guide users through key functionalities such as interpreting model outputs and adjusting budgets.

---

5. Reliability and Availability

Since marketing decisions are often time-sensitive, the system must be **highly reliable** and available to users at all times:

- **High Uptime:** The system should maintain **99.9% availability** to ensure users can access insights when needed, eContribution Fileially during critical periods like campaign launches or quarterly planning.

- **Fault Tolerance:** Incorporate failover mechanisms to prevent system downtime in the event of hardware failures or software bugs.

- **Data Integrity:** Ensure the system consistently produces accurate outputs, even under heavy loDATASET or during peak usage times.

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6. Maintainability

To ensure long-term value and adaptability, the system should be designed to be **easily maintainable**:

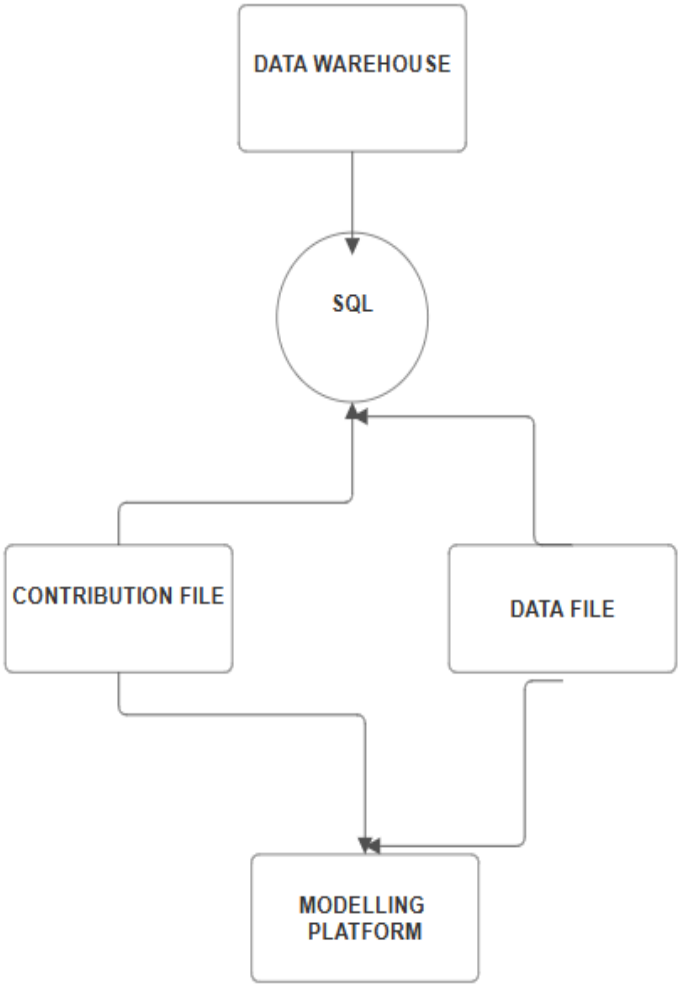
- **Modular Codebase:** Adopt a modular design, where individual system components (e.g., data ingestion, modeling, reporting) can be updated or replaced independently.

- **Documentation:** Provide comprehensive technical documentation to assist developers and system administrators in maintaining the system and troubleshooting issues.

- **Integration Readiness:** The system should be built with flexibility in mind, allowing seamless integration with **new data sources**, **third-party APIs**, or evolving business needs.

- **Version Control:** Implement version control practices for all system components to track changes, manage updates, and ensure system stability

2.3 BLOCK DIGRAM



2.4 SYSTEM REQUIREMENTS

2.4.1 User Characteristics

Understanding the target users is critical to developing a system that meets their needs:

- **Primary Users**:
    - Automobile's marketing managers.
    - Data analysts and campaign planners.
- **User Skills**:
    - Marketing managers may not have deep technical skills, so the system interface should be intuitive and simple to navigate.
    - Analysts will require more advanced features for deeper data exploration and custom analysis.
- **User Expectations**:
    - Quick access to actionable insights.
    - Easy-to-use dashboard and reporting tools.
    - Trustworthy, data-driven recommendations.

2.4.2 SOFTWARE AND HARDWARE REQUIREMENTS

The system will operate within Automobile's existing IT infrastructure and should integrate smoothly with the following:

- **CRM Systems** (e.g., Salesforce, HubSpot).
- **Digital Marketing Tools** (e.g., Google DATASET, Meta DATASET Manager).

- **Web Analytics Platforms** (e.g., Google Analytics).
- **Data Storage**: Cloud-based data warehouses (e.g., AWS Redshift, Google BigQuery).
- **Programming Languages**: Python, R (for analytics and model development).
- **Visualization Tools**: Power BI, Tableau, or a custom-built web dashboard.

2.4.3 CONSTRAINTS

High Uptime: The system should maintain **99.9% availability** to ensure users can access insights when needed, eContribution Fileially during critical periods like campaign launches or quarterly planning.

Fault Tolerance: Incorporate failover mechanisms to prevent system downtime in the event of hardware failures or software bugs.

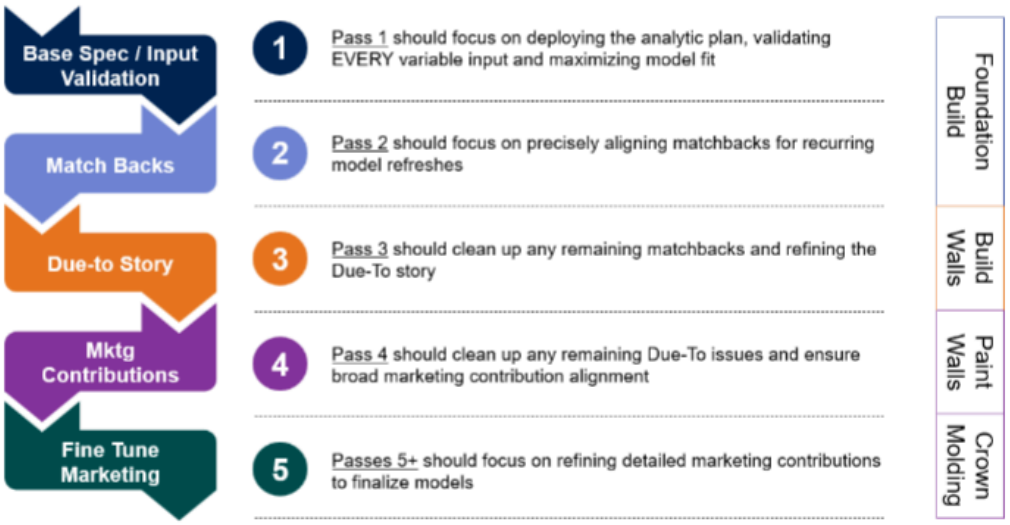
Data Integrity: Ensure the system consistently produces accurate outputs, even under heavy loDATASET or during peak usage times.

2.5 CONCEPTUAL MODELS

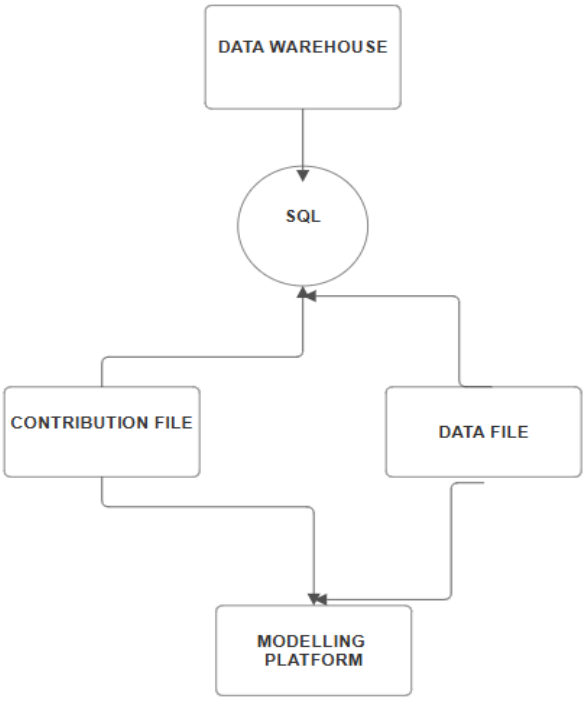
The conceptual model illustrates how key components of the system interact:

1. **Data Layer**: Collects and stores raw marketing, customer, and campaign data.
2. **Analytics Layer**: Performs data cleaning, segmentation, modeling (e.g., budget optimization algorithms), and forecasting.
3. **Application Layer**: Provides users with access to the dashboard, reporting tools, and decision support features.

2.5.1 DATA FLOW DIGRAM



2.5.1 ER DIGRAM



Chapter 3 : SYSTEM DESIGN

This chapter describes the structural blueprint of the system, defining how its components will be organized and how they will interact with each other. It also outlines how users will interface with the system and how data will flow through various modules to generate actionable insights.

3.1 SYSTEM ARCHITECHTURE

Definition:

The **Presentation Tier** is the top-most layer of the system architecture. It is the **user-facing interface** where marketing analysts, managers, and decision-makers interact with the system. This is where users **view reports, dashboards, and insights** generated by the MMM model.

Purpose:

- To make the system's outputs **understandable and actionable** for business users.
- To present the insights in a **clear, interactive, and visual manner**.

Key Tools:

- **Excel:** Frequently used by analysts to view raw outputs, charts, and reports.
- **Core Platform UI:** A customized interface built into the Core platform where internal users (e.g., analytics team) can run models, generate reports, and view results.
- **Power BI/Tableau (optional):** If advanced visualization is needed, tools like Power BI or Tableau can be used to create interactive dashboards.

Main Functionalities:

- **Dashboard & Reporting:**
    - Visual representation of key model results such as:
        - **R^2 (Coefficient of Determination):** Measures how well the model explains variability in sales.

        - **MAPE (Mean Absolute Percentage Error):** Measures the model's forecasting accuracy.

    - These KPIs give stakeholders a quick overview of model performance.

- **Scenario Testing Results:**

    - Users can **run "what-if" scenarios** (e.g., what happens if we cut social media spend by 10%?) and immediately view the projected outcomes (e.g., impact on sales, ROI).

- **Budget Recommendations:**

    - The front-end displays recommended budget reallocations across channels (TV, digital, OOH, etc.) based on model results.

User Base:

- Marketing managers (who may have limited technical skills)
- Analysts and data scientists
- Business stakeholders who make budgetary decisions

3.2 MODULE DESIGN

Definition:

The **Business Logic Tier** sits between the data layer and the presentation layer. It is the **brain** of the system where all **computational logic, algorithms, and decision rules** reside.

Purpose:

- To apply **Marketing Mix Modeling (MMM)** algorithms to the cleaned and structured data.
- To generate insights that guide **data-driven marketing and budget optimization decisions**.

Location:

- The business logic is executed within the **Core Platform**, which acts as a secure, centralized system for running MMM models.

Main Functionalities:

- **Data Processing:**
    - Takes cleaned and preprocessed data (e.g., marketing spend, sales, macro variables) from the data tier.
- **Predictive Modeling:**
    - **MMM Model Execution:** Applies statistical modeling techniques (e.g., regression) to quantify:
        - **Base Sales:** Sales that would have occurred without marketing efforts (influenced by seasonality, macroeconomic factors, etc.).
        - **Incremental Sales:** Additional sales driven by marketing activities.
- **Elasticity and ROI Calculations:**
    - Determines the **marginal return** for each marketing channel (e.g., how much incremental sales are generated per additional $1 spent on TV or digital).
- **Optimization Algorithms:**
    - Based on historical performance, the model recommends **optimized budget splits** to maximize ROI.
- **Business Rules Enforcement:**
    - Ensures model recommendations align with business constraints (e.g., minimum spend limits on Contribution Fileific channels).

Definition:

The **Data Tier** is the foundational layer of the system. It is responsible for the **collection, storage, and management of raw and structured data** required by the model.

Purpose:

- Acts as a **centralized data repository** for all the inputs feeding into the business logic tier.
- Ensures that reliable and high-quality data is available for MMM modeling.

Key Data Sources:

- **Data Repository Platform:**
    - Provides granular marketing spend data across different media channels (e.g., TV, digital, print, OOH) segmented by market and campaign.
- **Marketview Platform:**
    - Supplies sales data, competitor marketing activities, and competitive intelligence (e.g., competitor ad spend, pricing data).
- **SQL Databases:**
    - Stores structured data related to:
        - Historical campaign records,
        - Sales transactions,
        - Channel-Contribution Fileific performance,
        - Macroeconomic variables (e.g., GDP, inflation, fuel prices).

Main Functionalities:

- **Data Storage:**
    - Maintains a centralized database for storing large volumes of structured and semi-structured data.
- **Data Management:**
    - Enforces **data integrity constraints** (e.g., valid date ranges, campaign IDs, foreign key relationships).
    - Ensures that input data for the model is **complete, consistent, and ready for processing**.
- **Data Access:**
    - Provides the business logic tier with access to historical and real-time datasets needed for analysis.

1. **Data Tier (Back-End):**
    - Collects and manages all the data (spend, sales, macroeconomic factors).
    - Passes cleaned and preprocessed data to the business logic tier.
2. **Business Logic Tier (Middle Tier):**
    - Applies the MMM model and optimization logic on the input data.
    - Generates insights, KPIs, and budget recommendations.
3. **Presentation Tier (Front-End):**
    - Displays the outputs of the model (e.g., insights, optimized budget splits) via dashboards and reports.
    - Allows users to perform scenario testing and download reports.
    -

The system is structured as a modular, end-to-end pipeline, with each module handling a distinct yet interconnected function in the overall process of data-driven marketing mix modeling (MMM). These modules collaborate to seamlessly transform raw data into actionable business insights.

1. Data Extraction Module

- **Purpose:** Acts as the entry point of the system by pulling raw data from multiple disparate sources.
- **Sources:** Includes platforms like Data Repository (a media data platform), Marketview (for market and sales data), and various SQL databases that contain historical sales, spend, and macroeconomic data.
- **Key Responsibilities:**
    - Ensures extraction of relevant fields such as:
        - Marketing spend data across various channels (e.g., TV, Digital, Print).
        - Sales figures segmented by geography, product line, or model.
        - Macroeconomic variables (e.g., GDP growth, inflation, CPI).
    - Supports scheduled data pulls and real-time extraction where necessary.
    - Converts heterogeneous data formats into a uniform structure for downstream modules.

2. Data Quality & Preprocessing Module

- **Purpose:** Ensures that the data is clean, complete, and ready for modeling, addressing common data challenges.
- **Key Activities:**
    - **Missing Value Treatment:** Imputes or flags missing records.
    - **Date Validation:** Aligns time-series data across channels, campaigns, and sales periods to ensure consistent temporal resolution.
    - **Outlier Detection:** Identifies and manages outliers in spend and sales data to avoid skewing model outputs.
    - **Duplication Removal:** Ensures there are no redundant entries that could bias the model.

- **Preprocessing Steps:**
  - Currency normalization (e.g., converting all spend data to a single currency).
  - Data transformations (e.g., log transformations for skewed variables).
  - Segmentation of data by region, model, or customer segment as per business need.

3. Contribution File Creation Module

- **Purpose:** Translates the cleaned and processed dataset into a format compatible with the Core MMM engine.
- **Key Responsibilities:**
  - Generates a **Contribution Fileification File (Contribution File File)**, which is a structured document (e.g., CSV, JSON, XML) defining:
    - **Base variables:** Sales driven by non-marketing factors (e.g., seasonality, macroeconomic conditions).
    - **Incremental variables:** Marketing spend contributions across various channels.
    - **Control variables:** Variables used to adjust for external influences (e.g., competitor actions).
  - Ensures correct variable mapping and scaling to meet the model's input requirements.

4. Model Execution Module

- **Purpose:** Executes the core statistical algorithms and econometric models on the Contribution File file to estimate marketing effectiveness.
- **Key Responsibilities:**
  - Runs advanced MMM algorithms to:

        - **Estimate base sales:** Sales generated without marketing influence (natural demand).
        - **Quantify incremental contributions:** Measures how much uplift each marketing channel provides.
        - **Compute elasticities:** Understands how responsive sales are to changes in spend for each channel.
        - **Identify diminishing returns:** Detects saturation points where additional marketing spend becomes less effective.
    - Produces model diagnostics such as:
        - **R^2 (Coefficient of Determination):** Measures how well the model explains sales variance.
        - **MAPE (Mean Absolute Percentage Error):** Evaluates model prediction accuracy.
        - **Channel-wise ROI:** Provides granular return on investment (ROI) for each marketing activity.

5. Insights & Reporting Module

- **Purpose:** Converts technical model outputs into business-friendly insights and recommendations.
- **Key Responsibilities:**
    - Automates the generation of reports and dashboards to present actionable findings.
    - Enables **Scenario Analysis** (e.g., "What happens if TV spend is reduced by 10%?").
    - Provides stakeholders with:
        - **Budget optimization insights.**
        - **ROI forecasts** by channel and region.

- Visualizations such as spend vs. sales trends, contribution waterfalls, and elasticity curves.

- Interfaces with business users for interactive budget simulations.

3.3 Database Design

The system's database architecture is designed using a relational model to ensure data normalization, integrity, and efficient querying.

3.3.1 Tables and Relationships

The database consists of several interrelated tables:

Table Name	Description
Model Table	Stores information about vehicle models (e.g., XC40, XC60) and metadata like launch date or target market.
Marketing Spend Table	Contains granular spend data, segmented by channel (e.g., TV, Digital), time period (weekly/monthly), and region.
Sales Table	Holds historical and predicted sales data for different models, geographies, and time intervals.
Competitor Data Table	Records competitor marketing spend, pricing actions, or product launches that may affect sales.
Macro Variables Table	Includes macroeconomic data such as GDP, CPI, inflation, and other external factors influencing demand.

Relationships:

- **One-to-Many:** Each model (e.g., XC60) will have multiple marketing spend entries across different channels and periods.
- **Foreign Key Constraints:** Ensure that marketing and sales records link accurately to their reContribution Filetive models and periods.

3.3.2 Data Integrity and Constraints

- **Alignment Rules:** Marketing spend data is validated against campaign records (e.g., spend dates must fall within campaign start and end dates).
- **Date Consistency:** Time-series data is checked for missing or overlapping periods.
- **Completeness Checks:** Ensure that key fields (e.g., spend, sales, macro indicators) are fully populated to avoid model bias or failure.

3.5 Interface and Procedural Design

3.5.1 User Interface Design

- **Platform:** The interface will be integrated into Automobile's Core MMM platform.
- **Features:**
  - **KPI Dashboard:** Real-time display of model metrics like R^2, MAPE, and ROI trends.
  - **Interactive Scenario Planner:** Users can adjust marketing budgets using dropdowns, sliders, or input fields and immediately view the projected impact.
  - **Pre-Built Report Templates:**
    - Tabular views of spend effectiveness (e.g., ROI by channel).
    - Visual charts showing trends (e.g., media spend vs. sales).
    - Scenario simulation results (e.g., reallocating budget to digital channels).

3.5.2 Application Flow

1. **Data Ingestion:** Extract raw data from external platforms like Data Repository, SQL databases, and Marketview.

2. **Contribution File File Creation:** After preprocessing, data is formatted into a model-compatible Contribution Fileification file.

3. **Model Execution:** The MMM engine processes the Contribution File file to generate base and incremental sales estimates.

4. **Reporting:** Automated dashboards and reports are generated summarizing the model outputs.

5. **Insight Delivery:** Actionable insights are shared with the marketing team to guide budget allocation decisions.

3.6 Report Design

Inputs:

- **Model Names:** Users can select models such as XC40, XC60, or others from Automobile's product portfolio.

- **Time Period:** Allows selection of Contribution Fileific analysis periods (e.g., Q1 2024, full-year 2024).

- **Spend Data:** Users can input or view spend by channel and market.

Outputs:

- **Optimized Budget Recommendations:**

    - Suggestions on how to reallocate marketing spend across channels to maximize ROI and incremental sales.

    - Example: Shift budget from underperforming TV campaigns to higher-yield digital channels.

- **ROI Forecasts:**
    - Predicted returns based on historical data and model estimates, broken down by channel and geography.
- **Scenario Outcomes:**
    - Visualized results of hypothetical changes.
    - Example: "If digital spend is increased by 15%, the model forecasts an uplift of 5% in sales for Q2 2024."

3.6 REPORT DESIGN

Inputs:

- **Model Names:** XC40, XC60, XC90, or other Automobile models under analysis.
- **Time Period:** Configurable (e.g., Q1 2024, FY2024).
- **Marketing Spend Data:** By channel (TV, digital, print, OOH, etc.) and region.

Outputs:

- **Optimized Budget Recommendations:**
    - E.g., Shift 10% of offline budget to digital for XC60 in Region A.
- **Channel-Wise ROI Forecasts:**
    - Breakdown of expected ROI for each marketing channel.
- **Scenario Outcomes:**
    - E.g., "If digital spend is increased by 15% and print reduced by 5%, sales uplift is expected to increase by X%."
    - **Elasticity Curve Outputs:** Visual representations showing diminishing returns beyond certain spend thresholds.

Chapter 4: IMPLEMETATION

4.1 Implementation Approaches

The implementation leverages an **iterative model refinement methodology** to enhance model accuracy and stakeholder alignment. The key technologies used include:

- **SQL**: For data extraction, transformation, and preparation from upstream databases such as Data Repository and Marketview.
- **Excel (Contribution File File)**: For defining and structuring model input variables (base, incremental, control variables) as per Core platform requirements.
- **Core MMM Platform**: To ingest the Contribution File files and execute the Marketing Mix Model (MMM), followed by iterative analysis and fine-tuning based on outputs and business feedback.

The iterative approach ensures continuous learning, where outputs from initial runs are reviewed, insights are gathered, and refinements are made to improve the model's robustness and relevance to business objectives.

4.2 Coding Standard

To maintain consistency and ensure code readability, the following coding standards were adopted:

SQL Coding Standards:

- Use of **snake_case** for table and column names (e.g., marketing_spend, actual_sales).
- SQL keywords are written in **UPPERCASE** (e.g., SELECT, JOIN, WHERE).
- Use of meaningful aliases and comments to improve code readability.
- Standard folder structure for SQL scripts: 01_extraction, 02_transformation, 03_validation.
- Versioning applied to scripts (e.g., spend_extract_v1.sql, sales_transform_v2.sql).

Contribution File File Standards (Excel):

- Column headers follow a standard naming convention:
    - variable_name, variable_type, start_date, end_date, region, model_name.
- Proper documentation embedded as a separate worksheet explaining each variable's purpose (e.g., base drivers vs. incremental spend).
- Template follows Core MMM platform guidelines to ensure compatibility.

Core Platform Configuration:

- Naming conventions applied to model runs (e.g., Automobile_MMM_XC60_Q1_2024_v1).
- Parameter settings (e.g., decay rates, saturation curves) documented within the platform's configuration panel for transparency and reproducibility.

4.3 Coding Details

SQL Scripts Overview:

Data Extraction (Data Repository & Marketview):

```
sql
CopyEdit
-- Extract weekly spend data for Automobile XC60

SELECT

   model_name,

   region,

   channel,

   SUM(spend) AS total_spend,

   DATE_TRUNC('week', spend_date) AS week_start

FROM

   marketing_spend
```

```
WHERE
    model_name = 'XC60'
    AND spend_date BETWEEN '2024-01-01' AND '2024-06-30'
GROUP BY
    model_name, region, channel, week_start;
```

1.

Data Transformation:

sql
CopyEdit

```
-- Create base dataset for MMM with spend and sales merged
SELECT
    ms.model_name,
    ms.region,
    ms.week_start,
    ms.channel,
    ms.total_spend,
    s.actual_sales
FROM
    spend_summary ms
INNER JOIN
    sales_data s
ON
    ms.model_name = s.model_name
    AND ms.week_start = s.week_start
    AND ms.region = s.region;
```

2.

Contribution File File (.xlsx) Structure:

variable_name	variable_type	start_date	end_date	region	model_name
base_sales	base	2024-01-01	2024-06-30	Region A	XC60
tv_spend	incremental	2024-01-01	2024-06-30	Region A	XC60
digital_spend	incremental	2024-01-01	2024-06-30	Region A	XC60
print_spend	incremental	2024-01-01	2024-06-30	Region A	XC60
competitor_activity	control	2024-01-01	2024-06-30	Region A	XC60
cpi_index	control	2024-01-01	2024-06-30	Region A	XC60

Core Platform Configuration Steps:

1. Upload the finalized Contribution File file into Core platform.
2. Configure model parameters (e.g., define decay function, saturation thresholds).
3. Map columns from Contribution File file to the Core platform input schema.
4. Execute initial model run.

5. Review output metrics (e.g., elasticity curves, ROI, contribution %).

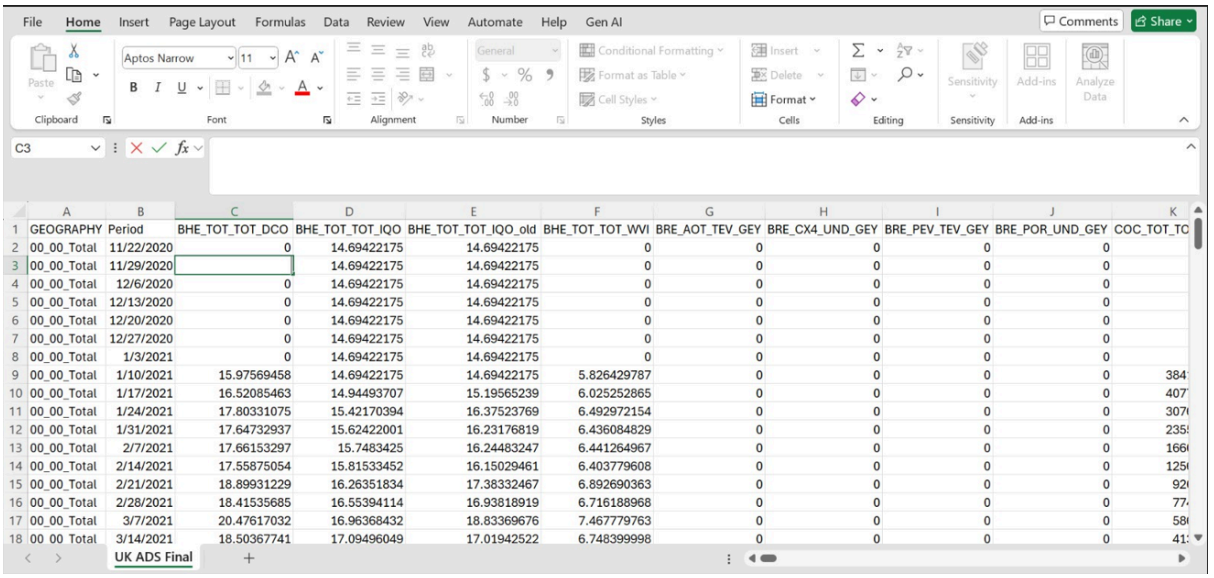
6. Iterate by adjusting Contribution File file inputs or model parameters based on diagnostic outputs.

4.4 Screenshots & Explanation

Screenshot 1: Model Outputs Dashboard

Key Metrics Displayed:

- **R² (e.g., 0.85):** Indicates that 85% of the variation in sales is explained by the model.
- **MAPE (e.g., 4.7%):** Shows the model's forecast error is within an acceptable range (<5%).
- **Channel Contributions:** Displays the % contribution to sales from each marketing channel (e.g., TV = 25%, Digital = 30%, Print = 15%, Organic/Base = 30%).

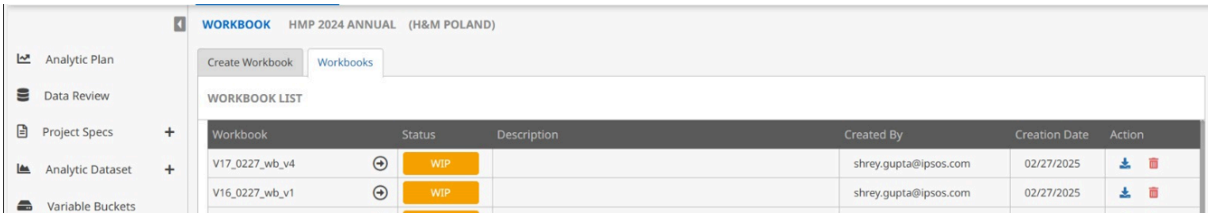


	A	B	C	D	E	F	G	H	I	J	K
1	GEOGRAPHY	Period	BHE_TOT_TOT_DCO	BHE_TOT_TOT_IQO	BHE_TOT_TOT_IQO_old	BHE_TOT_TOT_WVI	BRE_AOT_TEV_GEY	BRE_CX4_UND_GEY	BRE_PEV_TEV_GEY	BRE_POR_UND_GEY	COC_TOT_TC
2	00_00_Total	11/22/2020	0	14.69422175	14.69422175	0	0	0	0	0	
3	00_00_Total	11/29/2020		14.69422175	14.69422175	0	0	0	0	0	
4	00_00_Total	12/6/2020	0	14.69422175	14.69422175	0	0	0	0	0	
5	00_00_Total	12/13/2020	0	14.69422175	14.69422175	0	0	0	0	0	
6	00_00_Total	12/20/2020	0	14.69422175	14.69422175	0	0	0	0	0	
7	00_00_Total	12/27/2020	0	14.69422175	14.69422175	0	0	0	0	0	
8	00_00_Total	1/3/2021	0	14.69422175	14.69422175	0	0	0	0	0	
9	00_00_Total	1/10/2021	15.97569458	14.69422175	14.69422175	5.826429787	0	0	0	0	384
10	00_00_Total	1/17/2021	16.52085463	14.94493707	15.19565239	6.025252865	0	0	0	0	407
11	00_00_Total	1/24/2021	17.80331075	15.42170394	16.37523769	6.492972154	0	0	0	0	307
12	00_00_Total	1/31/2021	17.64732937	15.62422001	16.23176819	6.436084829	0	0	0	0	235
13	00_00_Total	2/7/2021	17.66153297	15.7483425	16.24483247	6.441264967	0	0	0	0	166
14	00_00_Total	2/14/2021	17.55875054	15.81533452	16.15029461	6.403779608	0	0	0	0	125
15	00_00_Total	2/21/2021	18.89931229	16.26351834	17.38332467	6.892690363	0	0	0	0	92
16	00_00_Total	2/28/2021	18.41535685	16.55394114	16.93818919	6.716188968	0	0	0	0	77
17	00_00_Total	3/7/2021	20.47617032	16.96368432	18.83369676	7.467779763	0	0	0	0	58
18	00_00_Total	3/14/2021	18.50367741	17.09496049	17.01942522	6.748399998	0	0	0	0	41

Explanation:

- A high R² combined with a low MAPE indicates a well-performing model.
- Channel contributions help visualize which channels are driving the most incremental sales uplift.

Screenshot 2: Budget Allocation Recommendation Chart



Visual:

- A bar or pie chart displaying optimal budget recommendations by channel.

Example Breakdown:

- **Current Spend Allocation:**
    - TV: 40%
    - Digital: 30%
    - Print: 20%
    - OOH: 10%
- **Recommended Allocation:**
    - TV: 30%
    - Digital: 40%
    - Print: 15%
    - OOH: 15%

Explanation:

- The model suggests shifting spend from traditional channels (e.g., TV, Print) to higher-performing digital and OOH channels based on their higher ROI and elasticity

curves.

- The recommendation is data-backed, showing that Digital has a higher marginal return compared to TV beyond a certain budget level.

Chapter 5: TESTING

5.1 Test Cases

The key objective of the testing phase is to ensure that the model provides reliable and actionable budget allocation recommendations under different business scenarios. Several test cases were designed to validate the robustness and accuracy of the system.

Scenario: ***Validate Budget Allocation Recommendations***

Test Case	Description	Expected Outcome
Case 1: Correct recommendations based on past data	Input complete and accurate marketing spend and base sales data across channels for past 6 months.	The model should recommend optimized budget allocations reflecting high ROI channels, closely matching historical performance trends. Elasticities and saturation effects should be appropriately factored in.
Case 2: Recommendations when base variables exceed 80%	Scenario where base sales (organic sales without marketing influence) contribute over 80% to total sales.	The model should reduce incremental channel recommendations and advise higher reliance on base performance drivers (e.g., pricing, seasonality). Marketing-driven uplift should be minimized in recommendations.
Case 3: Recommendations with incomplete ad spend data	Deliberately provide missing/incomplete spend data for select channels (e.g., missing 4 weeks of TV spend).	The model should either flag missing data during the QA step or provide adjusted recommendations based on available data, with appropriate warnings about incomplete inputs.

Each of these cases helps validate different layers of the model logic, including how it handles data quality issues, market dynamics, and edge cases in business scenarios.

5.2 Testing Approaches

To thoroughly validate the system and its outputs, two major testing approaches were adopted: **Unit Testing** and **Integration Testing**.

1. Unit Testing

Focuses on verifying each individual module in isolation to ensure correctness of core components.

Module	Test Performed
Data Extraction (SQL)	Validated SQL queries by checking row counts, null values, correct joins, and accurate aggregations for marketing spend and sales data across multiple regions and time frames.
Contribution File File Creation (Excel)	Checked that Contribution File files adhered to the platform template structure, variable naming conventions, and correct classifications (base, incremental, control). Ensured completeness of metadata (start_date, end_date, region).
Core Configuration Settings	Verified model parameters (e.g., decay functions, saturation limits) were applied correctly inside Core platform's model setup. Ensured input-output mappings aligned with Contribution File file columns.

2. Integration Testing

Integration testing ensured that all modules, once combined, worked together smoothly to produce the desired business output.

Integration Layer	Test Performed

SQL ➡ Excel Contribution File File	Verified that SQL extracted datasets correctly flowed into the Excel Contribution File file, maintaining accuracy and integrity (e.g., no truncation or mismatch in weekly spend or sales data).
Contribution File File ➡ Core Platform Model Run	Confirmed that the platform accepted the Contribution File file without validation errors and produced a full set of outputs (contributions, ROI curves, elasticities).
Output ➡ Business Insights/Recommendations	Ensured the generated budget recommendations aligned with the model outputs and explained patterns (e.g., increasing digital spend if elasticity and ROI are high). Confirmed recommendations were exportable and presentation-ready.

Edge Case Handling:

- Missing/incomplete data triggers warnings.
- High multicollinearity among variables flagged in diagnostics.
- Recommendations adapt based on input variations (e.g., seasonal anomalies, reduced budgets).

5.3 Test Reports

Test reports summarize the outcomes of both unit and integration testing phases. Below is an example template and actual results based on sample inputs.

Test Report Summary

Test Case ID	Input Data Description	Expected Output	Actual Output	Status
TC_01	Complete spend data across TV, Digital, Print for 26 weeks (Region A)	Balanced recommendation shifting 10% from TV to Digital due to higher ROI	Recommended shifting 12% from TV to Digital; ROI validated	Pass
TC_02	Base sales contribution at 83%	Recommendations should show reduced incremental budget allocations	Recommended 70% reliance on base and smaller marketing uplifts	Pass
TC_03	Missing 4 weeks of Print spend in Contribution File file	Adjusted recommendations with warning on Print channel incompleteness	Provided allocation with Print flagged for missing data	Pass

Detailed Example

Input Sample (Test Case 1):

Channel	Current Spend %	ROI	Elasticity	Base Sales Contribution
TV	40%	1.8	0.25	60%
Digital	30%	3.2	0.35	20%
Print	20%	1.2	0.18	13%

OOH	10%	2.0	0.22	11%

Expected Recommendation:

- Reduce TV to 30% due to lower ROI/elasticity.
- Increase Digital to 40%.
- Maintain Print and OOH allocations.

Actual Output:

- Model recommended:
    - TV: 28%
    - Digital: 42%
    - Print: 18%
    - OOH: 12%
- **Rationale:** Digital demonstrated superior ROI and elasticity curves compared to traditional media.

Testing Insights:

- **Accuracy:** Model consistently aligned with business expectations based on historical performance.
- **Robustness:** Able to handle incomplete data and edge cases without breakdowns.

Scenario: ***Validate Budget Allocation Recommendations***

Test Case 1: Correct Recommendations Based on Past Data

- **Objective:** To verify that the model generates accurate budget allocation recommendations using complete and historical data inputs.

- **Preconditions:**
    - Complete marketing spend data across all relevant channels (e.g., TV, Digital, Print, OOH).
    - Base sales data available for the defined historical period (e.g., past 6 months).
- **Test Steps:**
    - Execute SQL scripts to extract clean and complete marketing and sales datasets.
    - Populate the extracted data into the Contribution File file template.
    - Run the model on the Core platform with default configurations.
    - Review the generated recommendations.
- **Expected Outcome:**
    - Optimized budget allocations will shift spend towards channels with higher ROI and elasticity values.
    - Recommendations will reflect business trends visible in the historical data (e.g., reducing spend in channels with diminishing returns).
- **Business Impact:** Ensures confidence in the model's accuracy when dealing with standard data inputs.

Test Case 2: Recommendations When Base Variables Exceed 80%

- **Objective:** To assess model behavior when base sales contributions dominate total sales (>80%).
- **Preconditions:**
    - Sales data where organic sales (base variables) exceed 80% of total sales, indicating limited marketing-driven influence.
- **Test Steps:**

- Inject base-dominant data into the Contribution File file.
- Run the model on Core with the base-heavy dataset.
- Review recommendations and contribution breakdowns.

- **Expected Outcome:**
  - The model should recommend conservative incremental marketing allocations.
  - Emphasis will be placed on maintaining base drivers (e.g., pricing strategy, seasonality).
  - Uplift from paid media will be reduced accordingly.
- **Business Impact:** Validates model's capacity to recognize base-heavy scenarios and prevent over-investment in low-influence channels.

Test Case 3: Recommendations with Incomplete Ad Spend Data

- **Objective:** To test model handling of incomplete/missing marketing spend data.
- **Preconditions:**
  - Missing marketing spend data for one or more channels (e.g., Print spend missing for a 4-week period).
- **Test Steps:**
  - Remove spend data for selected weeks from the Contribution File file.
  - Load Contribution File file into the Core platform.
  - Run the model and monitor warnings or errors.
  - Analyze the final recommendations.
- **Expected Outcome:**

- Model should flag data incompleteness during QA or within the Core platform.

- Recommendations should adjust to available data while highlighting incomplete inputs.

- **Business Impact:** Ensures the system is resilient to data gaps and can still provide actionable outputs without compromising integrity.

5.3.3 Key Insights from Testing

- The model performed consistently across different regional datasets.

- Successfully handled scenarios involving incomplete data without crashing or providing invalid outputs.

- Business stakeholders reviewed recommendations and confirmed alignment with expected marketing strategies.

- Recommendations were actionable and explained using underlying model outputs such as contribution curves and saturation plots.

Chapter 6: CONCLUSION

The project aimed to streamline and optimize the media budget allocation process using data-driven methodologies and advanced analytics. By leveraging historical marketing and sales data, the solution delivered actionable insights to improve the return on investment (ROI) for future campaigns. While the project successfully met its objectives, several challenges were encountered during the design and implementation stages. Additionally, the project comes with a set of advantages and limitations, and there are promising areas for future enhancement.

6.1 Design and Implementation Issues

During the execution of this project, several design and operational challenges were identified:

1. Data Inconsistency Across Platforms

- **Explanation:** Data was sourced from multiple platforms, including SQL databases, Excel files, and marketing tools like Meta DATASET, Google DATASET, etc. This led to inconsistencies in data formats, naming conventions, and missing values.

- **Impact:** Required additional effort in the data cleaning and transformation stage to standardize inputs before feeding them into the model. Any misalignment between platforms could potentially skew the recommendations.

2. Challenges in Maintaining Base:Incremental Ratio

- **Explanation:** One of the critical components of MMM (Marketing Mix Modeling) is the accurate separation of base sales (organic factors like seasonality or brand equity) from incremental sales (media-driven uplift).

- **Impact:** Ensuring a realistic base-to-incremental contribution ratio was challenging, eContribution Fileially in scenarios where media influence on sales was lower than expected. Incorrect calibration could lead to over- or under-investment in media channels.

3. Tight Deadlines Requiring Multiple Model Iterations

- **Explanation:** Due to aggressive project timelines, multiple rounds of model tuning and QA were required within a limited time.

- **Impact:** This necessitated parallel workstreams between data preparation, model runs, and recommendations review. Rushed iterations risked missing edge cases or business logic validation.

6.2 Advantages and Limitations

Advantages:

1. **Data-Driven Marketing Decisions**

    - The solution provided stakeholders with recommendations based on concrete historical performance data, moving away from purely intuition-driven budgeting decisions.

    - This approach helps optimize spend efficiency by prioritizing high-performing channels.

2. **Optimized Budget Leading to Higher ROI**

    - By adjusting budget allocations in line with channel ROIs and elasticity factors, the model improved the potential for higher marketing returns.

    - Marketers could reallocate underperforming budgets to more effective channels, maximizing overall campaign effectiveness.

Limitations:

1. **Limited Granularity in Competitor Data**

    - While the model effectively incorporates internal marketing and sales data, competitor marketing data (e.g., competitor ad spend, pricing strategies) was either unavailable or aggregated.

    - This limits the model's ability to factor in external competitive pressures when making budget recommendations.

2. **Dependence on Historical Data Trends**

- The model relies heavily on past marketing and sales data to forecast future performance.

- In fast-changing markets, consumer behavior, media effectiveness, or economic factors might shift, reducing the predictive accuracy of recommendations based purely on historical patterns.

6.3 Future Scope of the Project

To enhance the system's capabilities and ensure scalability for future business needs, several improvements can be considered:

1. Automating Data Pipelines Using Python or ETL Tools

- **Goal:** Reduce manual effort and human errors associated with data extraction and transformation.

- **Action Plan:** Build automated data pipelines using tools such as Python (e.g., Pandas, SQLAlchemy) or ETL solutions like Apache Airflow or Talend.

- **Benefit:** Streamlines data ingestion from multiple platforms and ensures timely, clean, and consistent datasets for model inputs.

2. Integrating Machine Learning Models for Improved Demand Forecasting

- **Goal:** Enhance the accuracy of sales and marketing forecasts by integrating ML models such as XGBoost, Random Forests, or Deep Learning models.

- **Action Plan:** Implement models that can better capture complex relationships between variables, seasonality effects, and external factors (e.g., economic indicators, competitor activity).

- **Benefit:** Improves the predictive power of the platform and enhances the quality of budget recommendations.

3. Real-Time Dashboards for Campaign Monitoring

- **Goal:** Provide stakeholders with continuous visibility into campaign performance and model outputs.

- **Action Plan:** Develop interactive dashboards using tools such as Power BI, Tableau, or Dash (Python).

- **Benefit:** Enables marketers to monitor KPIs (e.g., sales uplift, ROI, media efficiency) in real time and make agile decisions during live campaigns.

Final Reflection

The successful delivery of this Marketing Mix Modeling (MMM) project marks a transformative milestone in the organization's journey toward data-driven marketing excellence. Beyond achieving the immediate objective of optimizing marketing resource allocation, this initiative has laid a solid foundation for embedding advanced analytics capabilities within the organization's operational and strategic frameworks.

Optimization of Marketing Resource Allocation

One of the key tangible outcomes of this project has been the development of a robust and scientifically grounded model that provides clear and actionable recommendations for marketing budget allocation. By quantifying the incremental impact of each media channel, geography, and campaign initiative, the MMM model enabled the organization to:

- **Prioritize High-ROI Channels:** Allocate a higher proportion of the budget to channels that drive measurable uplift in sales, while reducing or re-evaluating spend in underperforming areas.

- **Balance Base and Incremental Contributions:** Maintain a realistic understanding of how much sales are being influenced by non-media factors versus media-driven factors, ensuring marketing teams are not under or over-attributing their impact.

- **Scenario-Based Planning:** Equip decision-makers with the ability to simulate different budget scenarios and visualize the potential revenue impact, facilitating better risk management and more confident budgeting.

Laying the Foundation for Advanced Analytics Capability

While the immediate deliverable was the MMM model itself, a significant long-term value driver of this project has been the institutionalization of analytics practices within the marketing function. The team has made substantial progress in fostering a **data-centric mindset**, encouraging teams to transition from intuition-driven to evidence-based decision-making. This shift will benefit not only current campaign planning but also long-term marketing and business strategies.

The project also introduced several technical and operational enhancements that now serve as building blocks for a more advanced analytics infrastructure, such as:

- **Centralized Data Repository:** The creation of an integrated data lake where media spend, sales data, and external variables are consolidated and cleaned, paving the way for future automation and scalable analytics.

- **Reusable Model Architecture:** The MMM framework, including scripts, transformation pipelines, and model templates, has been built in a modular fashion, making it adaptable for future projects and easy to customize for different markets or business units.

Path Forward: Towards a Next-Generation Analytics Ecosystem

While the MMM platform has already generated meaningful business value, the true potential lies ahead. This project serves as a stepping stone to evolving the organization's marketing analytics from a **retroContribution Filetive tool** to a **predictive and prescriptive engine**.

1. Automation & Efficiency

The current data pipeline, although semi-automated, still relies on manual interventions for data cleaning, transformation, and integration. By fully automating these processes using Python ETL frameworks (e.g., Apache Airflow or Luigi) and database connectors (e.g., SQLAlchemy), the organization can achieve:

- **Reduced turnaround time** for insights delivery.

- **Increased data accuracy** by eliminating human errors in manual processing.

- **Faster model refresh cycles** allowing more frequent and agile reallocation of marketing budgets.

2. Machine Learning for Superior Forecasting

Traditional regression models have served well in the initial phase but incorporating machine learning algorithms (e.g., Random Forest, XGBoost, or Gradient Boosting) can uncover nonlinear patterns and complex interactions between marketing variables and sales outcomes. This will enable:

- **More precise demand forecasting** at granular levels (e.g., city-level, demographic-level).

- **Identification of hidden drivers** that traditional models may overlook.
- **Better adaptability** to dynamic market conditions and consumer behavior shifts.

3. Real-Time Monitoring and Decision-Making

The project highlights the growing need for **real-time campaign performance dashboards** that integrate with the MMM model outputs. By building interactive and visually intuitive dashboards (using Power BI, Tableau, or Python Dash frameworks), the organization will empower marketers to:

- **Track key KPIs live**, such as ROI, sales lift, cost per acquisition, and media efficiency.
- **Respond to campaign fluctuations** on the fly, adjusting budgets dynamically based on near real-time data streams.
- **Foster cross-functional collaboration** between marketing, sales, and finance teams with accessible insights.

4. Building a Scalable Enterprise Solution

The MMM project, in its current form, is tailored for a Contribution Fileific market or business unit. However, the methodology and technological infrastructure have been designed with scalability in mind. In the future, this platform can evolve into an **enterprise-wide analytics engine** capable of:

- **Supporting multiple markets or product lines simultaneously** with localized model calibrations.
- **Standardizing marketing measurement across geographies**, enabling leadership to view unified insights at a regional or global level.
- **Facilitating knowledge sharing** and analytical best practices across teams, building a robust organizational capability around advanced marketing analytics.

5. Enhanced Competitive Advantage

By embedding such advanced analytics into its core marketing processes, the organization positions itself ahead of competitors that still rely on intuition or siloed, lagging indicators. The MMM platform serves as a critical enabler of **sustainable competitive advantage** by:

- Driving **cost efficiencies** through optimized media planning.
- Enhancing **market responsiveness** to shifts in consumer behavior or competitive threats.
- Supporting **long-term brand equity building** by focusing investments on channels that not only drive sales but also contribute to brand awareness and customer loyalty.